

Assignment 2 Report

GR5398 26 Spring - FinRL-Trading Quantitative Trading Strategy Track

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Backtest Window: 2018-01-01 to 2025-12-31

1. Objective

This assignment implements the Adaptive Rotation Strategy and extends it using the QQQ rolling selection logic from Assignment 1. The goal is to reproduce the original Adaptive Rotation framework, build a modified version, and compare both strategies against QQQ and SPY benchmarks.

The comparison includes the Original Adaptive Rotation Strategy, the Modified Adaptive Rotation Strategy with QQQ rolling selection logic, QQQ benchmark, and SPY benchmark.

2. Strategy Design

The strategy uses four asset groups: Growth Tech (AAPL, MSFT, NVDA, META, AMZN, GOOGL, TSLA), Real Assets (XOM, CVX, FCX, BHP, GLD, SLV), Defensive Assets (TLT, IEF, XLU, XLV, IAU, SHY, UUP), and Default Assets (SPY, QQQ, IAU, XLU, XLV).

The economic intuition is that different assets lead in different market regimes. Growth stocks tend to perform better in risk-on bull markets, real assets can perform well during commodity or inflation cycles, and defensive assets can help reduce downside exposure during market stress.

3. Original Adaptive Rotation Strategy

For each asset group, I calculate a trailing 12-week Information Ratio relative to QQQ: $\text{Information Ratio} = \frac{\text{mean}(\text{excess return})}{\text{standard deviation}(\text{excess return})}$. This selection rule rewards groups with strong and consistent relative performance.

After the strongest group is selected, assets inside the group are ranked by residual momentum. The strategy rebalances weekly and selects the top two assets from the chosen group with equal weights.

4. Modified Adaptive Rotation Strategy

The modified strategy follows the same group selection, residual momentum ranking, weekly rebalancing, and equal-weight allocation process as the original strategy.

The main change is applied to the Growth Tech sleeve. When Growth Tech is selected as the active group, I apply a QQQ rolling selection pre-filter before residual momentum ranking.

The rolling filter uses 6-month momentum, 12-month momentum, a 4-week skip period, and 13-week volatility adjustment. $\text{Raw Momentum Score} = 0.5 \times 6\text{M Momentum} + 0.5 \times 12\text{M Momentum}$. $\text{Risk-Adjusted Score} = \text{Raw Momentum Score} / \text{Recent Volatility}$.

5. Backtest Methodology

The backtest period is from 2018-01-01 to 2025-12-31. Daily adjusted close prices are downloaded from Yahoo Finance. Weekly prices are built using Friday close prices.

The strategy makes decisions on weekly rebalance dates and carries the selected weights forward until the next rebalance. To avoid look-ahead bias, daily portfolio returns are calculated using lagged weights. Transaction costs are included using a 10 bps cost rate applied to portfolio turnover.

6. Backtest Results

Portfolio	Cumulative Return	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown
Original Adaptive Rotation	1683.62%	43.39%	36.71%	1.18	-49.08%
Modified Adaptive Rotation	1448.50%	40.88%	35.19%	1.16	-49.53%

QQQ	308.42%	19.25%	24.05%	0.80	-35.12%
SPY	187.48%	14.12%	19.46%	0.73	-33.72%

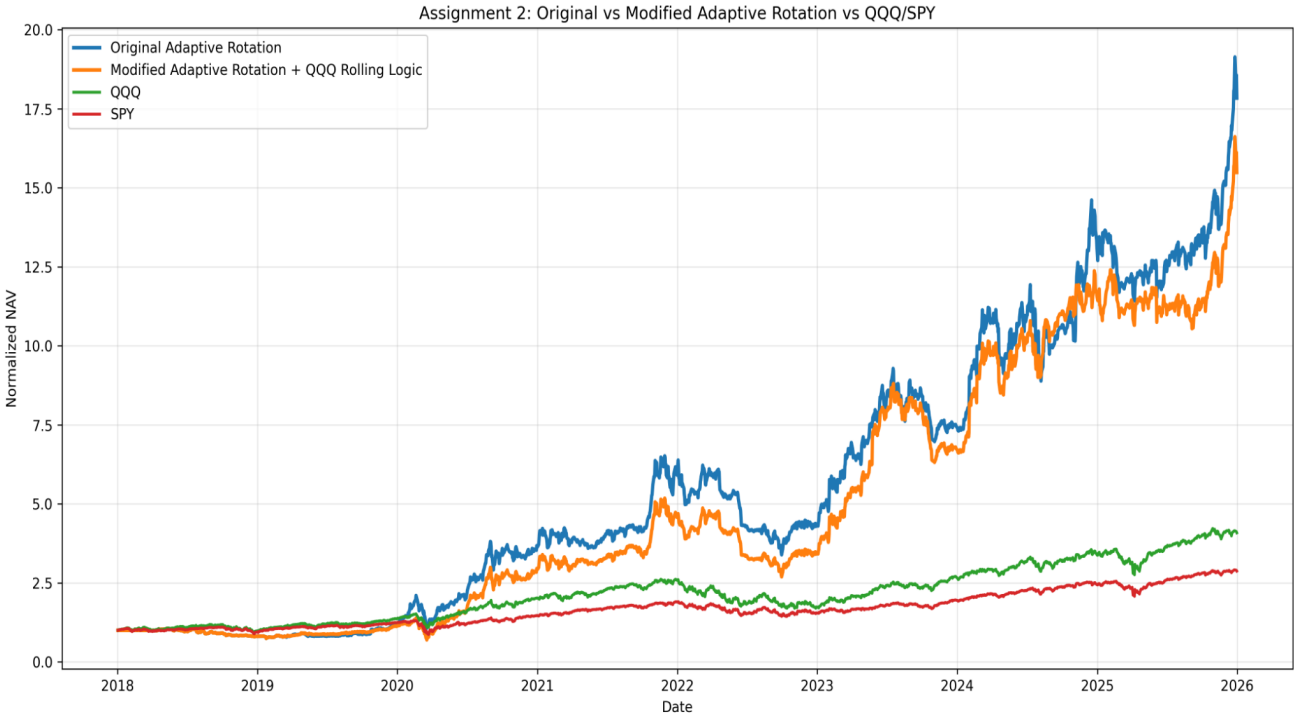


Figure 1. Original vs Modified Adaptive Rotation vs QQQ/SPY.

7. Result Analysis

The original Adaptive Rotation Strategy strongly outperformed QQQ and SPY during the 2018-2025 backtest period. It achieved a cumulative return of 1683.62% and an annualized return of 43.39%, compared with 308.42% and 19.25% for QQQ.

However, this return came with materially higher risk. The original strategy had annualized volatility of 36.71% and a maximum drawdown of -49.08%. This is more aggressive than the benchmarks because the strategy usually holds only two assets at a time.

The modified strategy achieved a cumulative return of 1448.50% and an annualized return of 40.88%. It did not outperform the original strategy in total return, but it slightly reduced annualized volatility from 36.71% to 35.19%.

8. Interpretation

The results show that the Adaptive Rotation framework is effective because it dynamically shifts capital toward stronger asset groups instead of holding a fixed portfolio. The original strategy performed best in terms of total return, likely because the concentrated top-two asset structure captured strong winners during the sample period.

The modified strategy was designed to improve stock selection inside the Growth Tech sleeve. The rolling momentum filter helped make the strategy more selective and slightly less volatile, but it did not improve absolute return. Therefore, the modified version should be interpreted as a conservative enhancement.

9. Limitations

The portfolio is highly concentrated because it holds only two assets after each rebalance. This can generate strong performance but also increases drawdown risk.

The QQQ rolling filter is only applied to the Growth Tech sleeve. When the selected group is Real Assets, Defensive Assets, or Default Assets, the modified strategy behaves the same as the original strategy.

The rolling filter uses fixed parameters, and the backtest uses adjusted close prices without fully modeling bid-ask spreads, market impact, or intraday execution timing.

10. Conclusion

This project implemented the original Adaptive Rotation Strategy and a modified version that integrates Assignment 1 QQQ rolling selection logic. The original strategy achieved the highest return, with cumulative return of 1683.62% and annualized return of 43.39%. The modified strategy achieved cumulative return of 1448.50% and annualized return of 40.88%.

The modified strategy did not outperform the original version, but it still significantly outperformed QQQ and SPY. It also slightly reduced volatility, suggesting that the QQQ rolling logic worked more as a risk-adjusted filter than as a pure return enhancer.

Future work could test the rolling filter across all asset groups, apply volatility targeting, or use a more explicit regime detection model.